



LOVNISH VERMA

Artificial Intelligence in Education

Concepts • Tools • Applications

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Session Overview

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What is Artificial Intelligence?

Definition, History & Key Concepts

02

Types of AI & Machine Learning

Narrow AI, ML, Deep Learning, Generative AI

03

AI Tools for Teachers

ChatGPT, Gemini, Canva AI, Grammarly, QuillBot & more

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AI in Education – Applications

Personalized learning, assessment, content creation

05

Responsible & Ethical AI Use

Bias, privacy, academic integrity

What is Artificial Intelligence?

"The science and engineering of making intelligent machines, especially intelligent computer programs that can think, learn, and act like humans."

Learn

Systems that improve from experience without being explicitly programmed

Reason

Analyze data, recognize patterns and make informed decisions

Act

Take actions to achieve goals — from answering questions to driving cars

Quick History: 1950 → 1990 → 2012 → 2022 → Today



Types of AI & Machine Learning

Artificial Intelligence (AI)

Machine Learning (ML) — learns from data without explicit rules

Deep Learning — neural networks inspired by the human brain

Generative AI — creates new content (text, images, audio)

Narrow AI

Designed for ONE specific task
(e.g. Face recognition, chess)

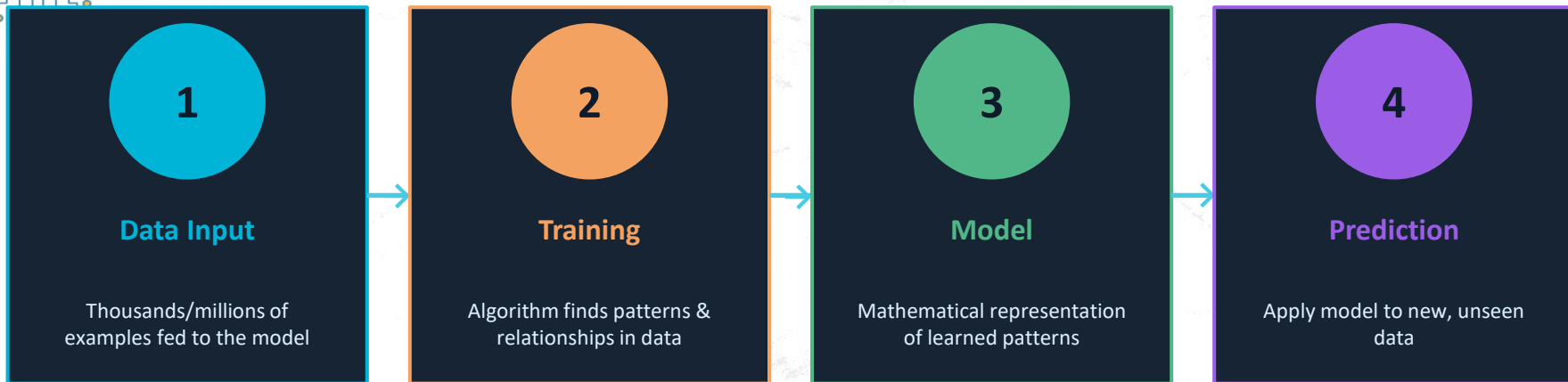
General AI

Human-level reasoning across all tasks
(still theoretical)

Super AI

Surpasses human intelligence (future concept)

How Does AI Learn?



Key Learning Approaches:

Supervised

Learns from labeled data
(e.g. spam vs not spam)

Unsupervised

Finds patterns in unlabeled data
(e.g. customer grouping)

Reinforcement

Learns by reward & penalty
(e.g. game-playing AI)

Popular AI Tools for Teachers

ChatGPT

OpenAI

- ◆ Lesson planning, Q&A, summaries, quiz generation

Google Gemini

Google

- ◆ Research assistant, document analysis, explanations

Canva AI

Canva

- ◆ Auto-generate presentations, posters & worksheets

Grammarly

Grammarly Inc.

- ◆ Grammar check, writing improvement, tone suggestions

QuillBot

QuillBot

- ◆ Paraphrasing, summarizing, citation generator

Socratic

Google

- ◆ Step-by-step homework help for students

ChatGPT for Teachers - Practical Use Cases



Lesson Planning

Prompt: "Create a 40-min lesson plan for Class 8 on Fractions with activities"



Quiz Generator

Prompt: "Generate 10 MCQs on Photosynthesis for Class 10 with answer key"



Simplify Content

Prompt: "Explain Newton's Laws of Motion in simple language for 12-year-olds"



Emails & Letters

Prompt: "Draft a parent communication regarding upcoming Annual Sports Day"



Translation Help

Prompt: "Translate this English passage to simple Punjabi for students"



Project Ideas

Prompt: "Suggest 5 science project ideas for Class 9 related to environment"



Tip: The more specific your prompt, the better the result!

AI Applications in Education



Personalized Learning

AI adapts content difficulty to each student's level. Khan Academy, Duolingo use this.



Intelligent Tutoring

AI tutors provide 24/7 help — explain concepts, answer doubts, give practice questions.



Auto Grading

Automatically grade MCQs, short answers & essays. Saves teacher time significantly.



Content Creation

Generate reading passages, worksheets, lesson notes & multimedia content instantly.



Learning Analytics

Track student progress, predict at-risk students, identify knowledge gaps.



Accessibility Tools

Text-to-speech, real-time captions, translation — making education inclusive.

AI in Action - Real World Examples

AI Around Us Every Day

-  Gmail spam filtering
-  Google Search predictions
-  Netflix/YouTube recommendations
-  Face unlock on smartphones
-  Siri, Alexa, Google Assistant
-  Face unlock on smartphones
-  Google Search predictions
-  Netflix/YouTube recommendations
-  Google Maps traffic prediction
-  Gmail spam filtering
-  Auto photo enhancement
-  Medical image diagnosis
-  Software Development (claude.ai)

IN AI in India

▶ DIKSHA Platform

AI-powered learning for school students

▶ PM e-Vidya

Digital education initiative with AI support

▶ Pratham AI

Personalized reading & math programs

▶ UPI & DigiLocker

AI fraud detection in fintech

▶ NPCI FASTag

AI-based toll & traffic management

▶ CoWIN Platform

AI for vaccine scheduling & analytics

Responsible & Ethical Use of AI



Bias in AI

AI can reflect and amplify human biases present in training data. Always question AI outputs.



Data Privacy

Never share students' personal details with AI tools. Read privacy policies carefully.



Academic Integrity

Using AI to write student assignments = cheating. Teach students to use AI as an aid, not a crutch.



Fact Checking

AI can 'hallucinate' — generate false information confidently. Always verify facts.

Remember: "AI is a tool — YOU are the teacher. Use it wisely!"

Hands-On Activity

Try it yourself - in pairs or individually!

Activity 1 (10 min)

Explore ChatGPT

- Go to chat.openai.com (free)
- Ask: 'Create a lesson plan on Water Cycle for Class 7'
- Edit the plan — is it good? What needs changing?
- Save the output and share with partner

Activity 2 (10 min)

AI Image Generation

- Go to canva.com → Text to Image
- Type: 'A diagram showing solar system for students'
- Download and discuss usefulness in classroom



Key Takeaways



- 1 AI is transforming every field, including education — teachers must stay ahead.
- 2 ML, Deep Learning & Generative AI are key sub-fields shaping today's AI tools.
- 3 Tools like **ChatGPT**, **Gemini** & **Canva AI** can save hours of teacher prep time.
- 4 AI in education enables personalized learning, smart assessments & analytics.
- 5 Always verify AI outputs, protect student data, and use AI ethically.

▶▶ Next: Session 4 — Demonstration / Hands-On Practice | Tea Break: 3:15–3:30 PM



Thank You!

Questions & Discussion

NIELIT Ropar

Deemed to be University

AI is not about replacing teachers — it's about empowering them!